

Способы связаться

yessenovmurat@gmail.com

www.linkedin.com/in/yessenov
(LinkedIn)

scholar.google.com/citations
(Portfolio)

[www.researchgate.net/profile/
Murat_Yessenov](http://www.researchgate.net/profile/Murat_Yessenov) (Portfolio)

ОСНОВНЫЕ НАВЫКИ

Community Outreach

Social Media Communications

Association Management

Languages

Russian (Native or Bilingual)

Kazakh (Native or Bilingual)

English (Professional Working)

Honors-Awards

Emil Wolf Outstanding Student
Paper

The Laser Technology, Engineering
and Applications Scholarship

Boris P. Stoicheff Memorial
Scholarship

Student of the Year

Murat Yessenov, PhD

Postdoctoral Fellow @Harvard University | PhD in Optics and
Photonics

Кембридж, Массачусетс, Соединенные Штаты Америки

Общие сведения

Optical scientist with a wide-ranging interest in exploring light, light-matter interactions, and their applications across diverse industries. I am currently a member of the Capasso Group at Harvard University, where I focus on space-time metasurfaces for nonlinear and quantum optics. My academic journey began at Nazarbayev University, where I studied physics, followed by a PhD in Optics and Photonics from CREOL, the University of Central Florida. In CREOL, I've had the privilege of working alongside talented colleagues on various projects in optics and photonics.

My research spans areas such as spectral analysis, pulse shaping through chirped Bragg gratings, and investigating particle acceleration using structured light sources. Studying the relativistic transformations of complex fields has been a particularly rewarding challenge. Throughout my career, I have enjoyed supporting and mentoring graduate and undergraduate students while engaging in collaborative projects with research groups from Harvard, Stanford, Tsukuba, and other esteemed institutions.

Participating in a team committed to scientific growth has been deeply fulfilling, particularly as a co-founder of initiatives aimed at advancing our field. I am eager to continue learning and collaborating with others, driven by a shared passion for advancing knowledge in optics and photonics. Please feel free to connect with me—I always welcome the opportunity to discuss new ideas and potential collaborations.

Опыт работы

Harvard University

Postdoctoral Fellow

февраля 2025 - Present (1 год 5 месяцев)

Cambridge, Massachusetts, United States

CREOL, The College of Optics & Photonics

4 года 2 месяца

Visiting Researcher

февраля 2025 - Present (1 год 5 месяцев)

Orlando, Florida, United States

Postdoctoral Researcher

мая 2022 - февраля 2025 (2 года 10 месяцев)

Worked on diverse projects including spectral analysis and pulse shaping via photographically recorded chirped Bragg gratings, particle acceleration with structured light

sources, and relativistic transformations of complex fields.

- Supervised graduate and undergraduate students actively involving research metrics and goals,

maintaining organization of the group progress, and guided in their research projects.

- Led collaborative projects in conjunction with research groups at Harvard University, Stanford University, Tsukuba University, and other institutions.

University of Central Florida

Graduate Research Assistant

августа 2017 - апреля 2022 (4 года 9 месяцев)

Orlando, Florida Area

- Worked on newly developed spatio-temporally shaped light waves.

- Developed theoretical models for free space propagation and nonlinear interaction

of finite-energy spatio-temporally structured light fields. Created 20+ comprehensive

optical simulations for modeling, design, DAQ, and data analysis that continue to be

used by students in the group.

- Participated in collaborative work with several groups within the institution and beyond.

- Published over 30 papers, with 12 as first author; presented at over 15 conferences

- Contributed to grant proposals for MURI, ONR, and Keck Foundation, and research articles in high-impact journals (Physical Review Letters, Nature Photonics, Nature Communications, Optica, etc.)

Education Center "Ziyatker"

Founder & CEO

сентября 2015 - мая 2017 (1 год 9 месяцев)

Astana

Founded and managed a center of supplemental education open for students of

all ages, with more than 10 different courses offered in STEM and English language.

- Official organizers of the regional round of the international math competition "Mathematics Without Borders" in 2016 and 2017, with over 1500 total participants in attendance.

- Within the second year of founding reached 250+ students/semester with 30 teachers employed, resulting in a cash flow of \$50,000.

University of Crete

Summer Intern

июня 2016 - июля 2016 (2 месяца)

Heraklion, Greece

Research internship at Crete Center for Quantum Complexity and Nanotechnology. I was involved in the simulation of quantum Ising models under the supervision of Dr. Navez and Dr. Tsironis

Nazarbayev University

Undergraduate Research Assistant

мая 2014 - мая 2015 (1 год 1 месяц)

Astana, Kazakhstan

During my undergraduate studies, I was involved in experimental research on elementary particle physics. Namely, I was a part of the team working on high-altitude Horizon-T systems - a system that measures global cosmic ray flux and local radiation background, located at the Tien Shan Mountain Cosmic Ray Station.

High School "Zerde"

Physics Teacher

сентября 2013 - мая 2014 (9 месяцев)

Astana, Kazakhstan

Prepared a group of high school students (4-5 students) for regional and national physics tournaments. Students won several regional olympiads and one National Olympiad.

Weizmann Institute of Science

Intern

июля 2013 - августа 2013 (2 месяца)

Rehovot Area, Israel

Intern at the Plasma Lab of Weizmann Institute of Science and an attendee of the International Summer Science Institute (ISSI) - a one-month program for 80 high-school students from 30+ countries around the world. During the one-month program, I conducted research in plasma physics under the supervision of PhD students and postdocs. By the end of the program, we presented our results in front of all participants.

Образование

University of Central Florida

Doctor of Philosophy - PhD, Optics/Optical Sciences · (2017 - мая 2022)

Nazarbayev University

Bachelor of Science - BS, Physics · (2013 - 2017)